

## Preliminary work.

This book assumes that you are already familiar with a number of mathematical ideas from your mathematical studies in earlier years.

This section outlines the ideas which are of particular relevance to Unit Four of the Mathematics Applications course and for which familiarity will be assumed, or for which the brief explanation given here may be sufficient to bring your understanding of the concept up to the necessary level.

Read this "preliminary work" section and if anything is not familiar to you, and you don't understand the brief mention or explanation given here, you may need to do some further reading to bring your understanding of those concepts up to an appropriate level for this unit. (If you do understand the work but feel somewhat "rusty" with regards to applying the ideas some of the chapters afford further opportunities for revision as do some of the questions in the miscellaneous exercises at the end of chapters.)

- ☞ Chapters in this book will continue some of the topics from this preliminary work by building on the assumed familiarity with the work.
- ☞ The miscellaneous exercises that feature at the end of each chapter may include questions requiring an understanding of the topics briefly explained here.

- **Number.**

The understanding and appropriate use of the rule of order, fractions, decimals, percentages, rounding, truncation, square roots and cube roots, numbers expressed with positive integer powers, e.g.  $2^3$ ,  $5^2$ ,  $2^5$  etc, expressing numbers in standard form, e.g.  $2.3 \times 10^4$  ( $= 23\,000$ ),  $5.43 \times 10^{-7}$  ( $= 0.000\,000\,543$ ), also called scientific notation, and familiarity with the symbols  $>$ ,  $\geq$ ,  $<$ , and  $\leq$  is assumed.

- **Percentages.**

It is assumed that you are able to express one quantity as a percentage of another quantity and find a percentage of a quantity. It is also assumed that you are able to increase or decrease an amount by a certain percentage by appropriate multiplication. For example: To increase an amount by 25% we multiply by 1.25.

To decrease an amount by 5% we multiply by 0.95.

Typical percentage questions:

- I Express 28 out of 40 as a percentage.

$$\begin{array}{lcl} \text{First express as a fraction:} & \frac{28}{40} & \\ \text{Then multiply by 100:} & \frac{28}{40} \times 100 & \\ & = & 70\% \end{array}$$

Thus 28 out of 40 is 70%.

II Find 28% of \$40.

Either: Divide by 100 to find 1%.

$$\frac{\$40}{100}$$

Then multiply by 28 to find 28%.

$$\frac{\$40}{100} \times 28$$

$$= \$11.20$$

Or: Use the decimal equivalent of 28%

$$28\% \text{ of } \$40 = \$40 \times 0.28$$

$$= \$11.20$$

The following examples of "typical percentage questions" will only show the "decimal equivalent" method.

III Increase \$40 by 28%.

$$\begin{aligned} \$40 \text{ increased by } 28\% &= \$40 \times 1.28 \\ &= \$51.20 \end{aligned}$$

IV Decrease \$40 by 28%

$$\begin{aligned} \$40 \text{ decreased by } 28\% &= \$40 \times 0.72 \\ &= \$28.80 \end{aligned}$$

IV 28% of an amount is \$40. Find the amount.

$$\begin{aligned} 28\% \text{ of an amount is } \$40 \\ 100\% \text{ of the amount} &= \frac{\$40}{0.28} \\ &= \$142.86 \text{ (nearest cent).} \end{aligned}$$

- **Straight line graphs.**

If points lie in a straight line their coordinates,  $(x, y)$ , obey a rule of the form

$$y = mx + c$$

and the relationship between  $x$  and  $y$  is said to be *linear*.

In this rule  $m$  is the gradient of the straight line and the point  $(0, c)$  is where the line cuts the vertical axis.

Hence the line  $y = 2x + 1$  has a gradient of 2 and cuts the  $y$ -axis at  $(0, 1)$ .

The line  $y = -x + 5$  has gradient of  $-1$  and cuts the  $y$ -axis at  $(0, 5)$ .

In the equation  $y = mx + c$  the value of  $m$ , the gradient, tells us the amount by which  $y$  increases for each unit increase in  $x$ .

- Solving equations.**

It is assumed that you are already familiar with the idea that solving an equation involves finding the value(s) the unknown can take that make the equation true. For example,  $x = 5.5$  is the solution to the equation  $15 - 2x = 4$  because  $15 - 2(5.5) = 4$ .

To solve an equation, eg  $15 - 2x = 4$ , we could proceed *mentally*:

We know that fifteen take *eleven* equals four.

Thus  $2x = 11$  and so  $x = 5.5$ .

We could use the *solve facility* of some calculators.

Or we could use a *step by step approach* to isolate  $x$ :

|                                                       |                      |
|-------------------------------------------------------|----------------------|
| Given the equation:                                   | $15 - 2x = 4$        |
| Add $2x$ to both sides to make the $x$ term positive: | $15 = 4 + 2x$        |
| Subtract 4 from both sides to isolate $2x$ :          | $15 - 4 = 2x$        |
|                                                       | $\therefore 11 = 2x$ |
| Divide both sides by 2 to isolate $x$ :               | $5.5 = x$            |
|                                                       | $\therefore x = 5.5$ |

It is also anticipated that you are able to solve more involved linear equations, for example:

$$5x - 7 - 2x - 8 = 30 - 6x \quad \text{Solution: } x = 5$$

$$\frac{x+3}{2} - 1 = 5 \quad \text{Solution: } x = 9$$

$$2(x+3) - 3(2x+1) = -5 \quad \text{Solution: } x = 2$$

and simultaneous linear equations, for example:

|           |                                                        |                                                         |                                                         |
|-----------|--------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
|           | $\begin{cases} 3x + 2y = 11 \\ x + 2y = 1 \end{cases}$ | $\begin{cases} 5x - 2y = 6 \\ 3x + 2y = 26 \end{cases}$ | $\begin{cases} 2x + 3y = 12 \\ x + 4y = 11 \end{cases}$ |
| Solutions | $x = 5, y = -2$                                        | $x = 4, y = 7$                                          | $x = 3, y = 2$                                          |

In this unit we may also encounter equations involving exponents (powers), for example

$$x^4 = 81$$

$$2^x = 64$$

$$2^x = 11$$

Solve these mentally if the numbers involved allow it, otherwise use the solve facility of some calculators.

$$x^4 = 81$$

$$x = 3$$

$$2^x = 64$$

$$x = 6$$

$$2^x = 11$$

$$\text{solve}(2^x = 11, x)$$

$$\{x = 3.459431619\}$$

$$x = 3.46 \text{ (correct to 2 dp.)}$$

- **Simple and compound interest:**

Suppose we were to invest \$500 for 3 years in an account paying interest at the rate of 10% per annum.

Considering a **simple interest** approach in which the same amount of interest is added each year:

$$\begin{aligned}\text{Value after 3 years} &= \$500 + 3 \times 10\% \text{ of } \$500 \\ &= \$500 + 3 \times \$50 \\ &= \$650\end{aligned}$$

Alternatively, considering **compound interest**, in which the interest earned in one year itself earns interest in the next year:

$$\begin{aligned}\text{Value after 1 year} &= \$500 + 10\% \text{ of } \$500 \\ &= \$500 \times 1.1 \\ &= \$550 \\ \text{Value after 2 years} &= \$550 \times 1.1 \quad (\text{i.e. } \$500 \times 1.1^2) \\ &= \$605 \\ \text{Value after 3 years} &= \$500 \times 1.1^3 \\ &= \$665.50\end{aligned}$$

If we want the value after  $n$  years:

$$\text{Value after } n \text{ years} = \text{Initial value} \times 1.1^n$$

- **Recursion.**

It is assumed that you are familiar with the idea of defining a sequence by a **recursive rule** or **recurrence relationship**, i.e. a rule which tells us how the terms of the sequence recur. For example, the following rule

$$T_1 = 5, \quad T_{n+1} = T_n + 3 \quad (\text{or } T_1 = 5, \quad T_n = T_{n-1} + 3)$$

tells us that the first term of the sequence is 5 and each term after that is obtained by adding 3 to the previous term

Hence the sequence is: 5, 8, 11, 14, 17, 20, 23, ....

With adjacent terms having a common difference this is an example of an **arithmetic sequence**, also called an **arithmetic progression** or **AP**.

Thus all arithmetic sequences are of the form:

$$a, \quad a + d, \quad a + 2d, \quad a + 3d, \quad a + 4d, \quad a + 5d, \quad a + 6d, \quad \dots$$

In this general form we have a first term of "a" and **common difference** "d".

The  $n^{\text{th}}$  term is then given by  $T_n = a + (n - 1)d$ .

Sequences which progress such that each term of the sequence is the previous term multiplied by some constant number are said to be **geometric sequences**, **geometric progressions**, or **GPs**.

Thus all GPs are of the form:

$$a, \quad ar, \quad ar^2, \quad ar^3, \quad ar^4, \quad ar^5, \quad ar^6, \quad \dots$$

In this general form we have a first term of "a" and **common ratio** "r".

The  $n^{\text{th}}$  term is then given by  $T_n = a \times r^{n-1}$ .

GPs will have a recursive rule of the form  $T_1 = a, \quad T_{n+1} = r \times T_n$ .

It is also assumed that you are familiar with recurrence rules of the form:

$$T_1 = a, \quad T_{n+1} = b \times T_n + c.$$

This type of recursive rule will generate sequences for which each term, after the first, is obtained from the previous term by multiplying by some constant value,  $b$ , and then adding a constant value,  $c$ .

For example  $T_1 = 4000$ ,  $T_{n+1} = 1.01 \times T_n - 100$ .

Thus  $T_1 = 4000$   $T_2 = 1.01 \times T_1 - 100$   
 $= 1.01 \times 4000 - 100$   
 $= 3940$

$$T_3 = 1.01 \times T_2 - 100$$

$$= 1.01 \times 3940 - 100$$

$$= 3879.4$$

$$T_4 = 1.01 \times T_3 - 100$$

$$= 1.01 \times 3879.4 - 100$$

$$= 3818.194$$

etc.

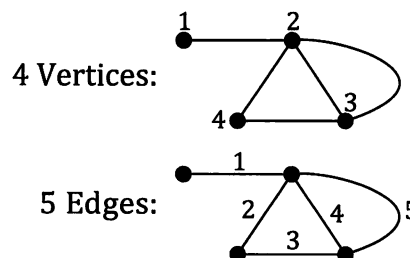
Recurrence relations of this form are particularly relevant to this unit.

- **Graphs or networks.**

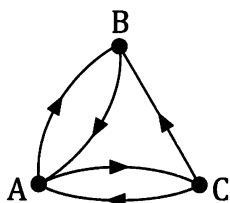
In mathematics a network or graph is a set of points, called **vertices**, connected by a set of lines, called **edges**.

☞ The junctions and endpoints are called **vertices** (singular: **vertex**).

☞ The connections are called **edges**.



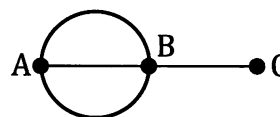
Graphs may be directed or undirected.



A directed graph

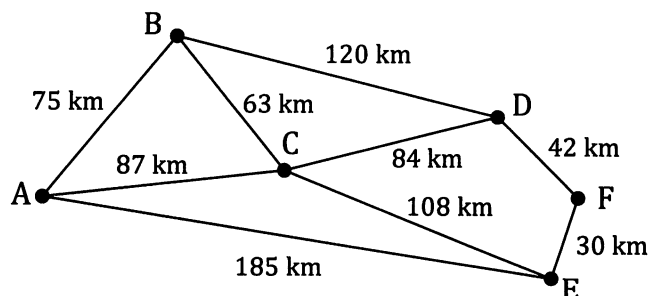
(Also referred to as a digraph)

The **directed edges** are sometimes referred to as **arcs**.



An undirected graph.

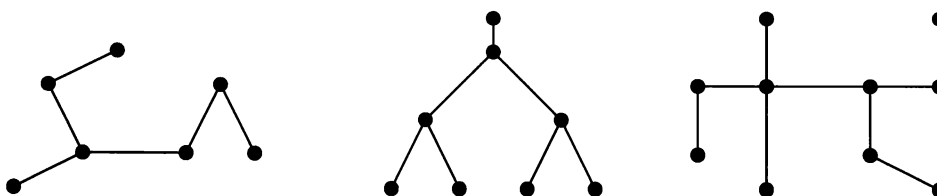
Some graphs show "weights" on each edge, perhaps distance, cost, time etc. Such graphs are referred to as weighted graphs. For example:



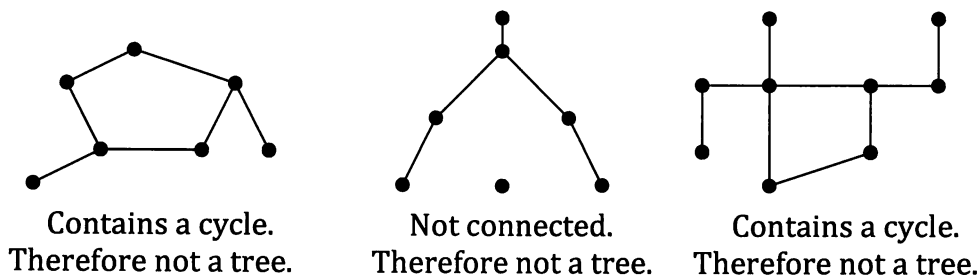
Whilst it is anticipated that from your study of *Unit Three of Mathematics Applications* you are familiar with many of the terms associated with graph theory (eg multiple edges, adjacent vertices, simple graphs, connected graphs, planar graphs, subgraphs, bridges, walks, paths, cycles etc,) in this unit we will be mainly focusing on analysing weighted graphs and in one such aspect an understanding of the word **tree** in the context of graph theory is important, and in another aspect **bipartite** graphs are significant. These two terms in particular are further explained below:

Any connected simple graph that contains no cycles is called a tree:

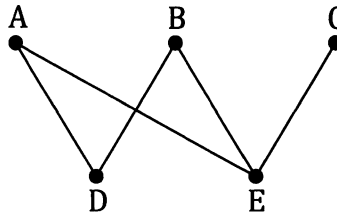
The following graphs are trees:



The following are not trees:



Consider the following graph:



Notice that if we consider the five vertices A, B, C, D and E as being in two sets:

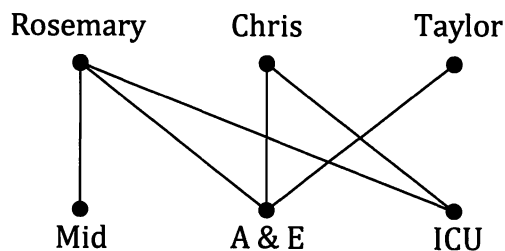
$\{A, B, C\}$  and  $\{D, E\}$

then every one of the five edges joins a vertex from one set to a vertex of the other set.

AD, AE, BD, BE, CE.

A graph in which the vertices can be split into two groups such that every edge joins a vertex from one group to a vertex of the other group, i.e. no edges join vertices from the same group, is called a **bipartite** graph. Thus, in a bipartite graph, every pair of adjacent vertices involves one vertex from one group and the other vertex from the other group.

For example, the bipartite graph below shows which of the three areas, midwifery (Mid), accident and emergency (A & E), and the intensive care unit (ICU), three nurses are qualified to work in.



- **Bivariate data.**

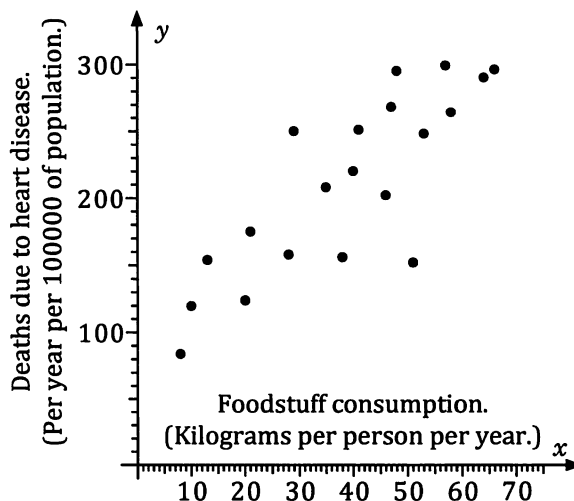
If a situation involves two data sets, for example the heights and weights of a number of people, we have **bivariate data**. We might then be interested in investigating whether there is an **association** between the data sets, for example is there some association between a child's age and their foot size? Do older children generally have bigger feet? Does a child's age in any way explain their foot size? In this case age would be the **explanatory variable**, also called the independent or predictor variable. Foot size would be the **response variable**, also called the dependent or predicted variable.

- Regression.**

The scattergraph on the right shows the consumption of a particular food commodity on the horizontal  $x$ -axis, and heart disease death rate on the vertical  $y$ -axis, for twenty countries.

The location of the points suggests that as one of the measures increases then the other does too.

This apparent **correlation** between the two variables could be modelled by drawing a **line of best fit** or **trend line**, as shown on the next graph.

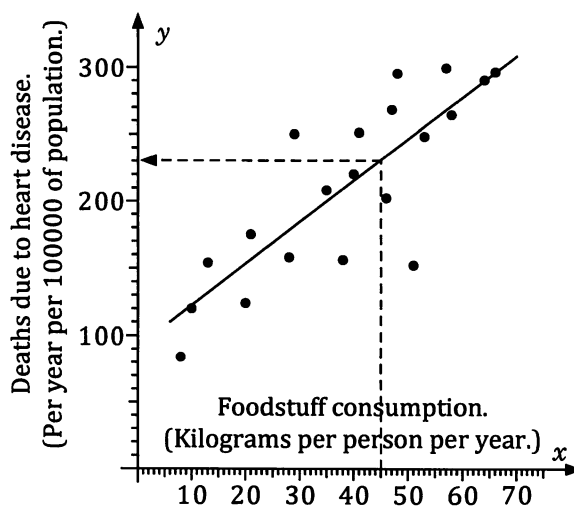


With one variable generally increasing as the other increases this line of best fit has a positive gradient.

We say that a **positive linear correlation** seems to exist.

The line of best fit allows predicted values to be suggested.

For example we could suggest that a country with an intake of the foodstuff of 45 kilograms per person per year would have a heart disease death rate of approximately 230 persons per year per 100000 of population.



Rather than having to draw this "line of best fit" by eye we can obtain its equation and predict values using the **linear regression** techniques of some calculators and spreadsheets.

The display on the right gives the equation of the line of best fit, or **least squares regression line**, as

$$y = 3.10x + 90.9.$$

Linear Reg

$$a = 3.09976491$$

$$b = 90.894086$$

$$r = 0.82280944$$

$$r^2 = 0.67701539$$

$$y = ax + b$$



Given that the data for the previous scattergraph is as shown below **make sure that you can obtain this equation from your calculator.**

|     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| $x$ | 57  | 20  | 13  | 38  | 51  | 48  | 58  | 35  | 29  | 8  | 28  | 21  | 10  | 40  | 41  | 46  | 53  | 64  | 66  | 47  |
| $y$ | 299 | 124 | 154 | 156 | 152 | 295 | 264 | 208 | 250 | 84 | 158 | 175 | 120 | 220 | 251 | 202 | 248 | 290 | 296 | 268 |

We can then calculate (or determine automatically from our calculator) a predicted value for " $y$ " (in this case deaths per year per 100 000 of population) for an " $x$ " value (in this case foodstuff consumption) of 45.

$$y = 3.0998(45) + 90.8941 \text{ i.e. approximately } 230.$$

The " $r$ " given in the previous calculator display is called **Pearson's correlation coefficient**. It will lie in the interval  $-1 \leq r \leq +1$ . This correlation coefficient is a measure of the strength and nature of the linear association, or correlation, between the variables. It informs us how closely the relationship between two sets of data approximates to a linear relationship.

A correlation coefficient of  $-1$  indicates a perfect linear relationship with a negative gradient (as one variable increases the other decreases) and a correlation coefficient of  $+1$  indicates a perfect linear relationship with a positive gradient (as one variable increases the other increases).

(The display also shows  $r^2$ , the **coefficient of determination**. This gives the proportion of the variation that can be explained by the linear relationship. I.e. in the previous situation the  $r^2$  value of  $0.677$  means that approximately 68% of the variation in death rates due to heart disease can be explained by, or accounted for, by the variation in the foodstuff consumption.)

If there is an association between the two variables then as one changes we should expect the other to change, but that is not to say that the changes in one variable *cause* the changes in the other. There could be other factors at work.

If we use the fact that a strong correlation seems to exist, to predict a " $y$ -value" for some " $x$ -value", we can be more confident of this prediction if **interpolation** is involved, i.e. if the  $x$ -value used is within the range of  $x$ -values covered by the given data.

If instead we are **extrapolating**, i.e. using an  $x$ -value outside the range covered by the  $x$ -values in the given data, we would be less confident with any prediction made, our confidence decreasing the further the  $x$ -value is from the range of  $x$ -values in the given data.

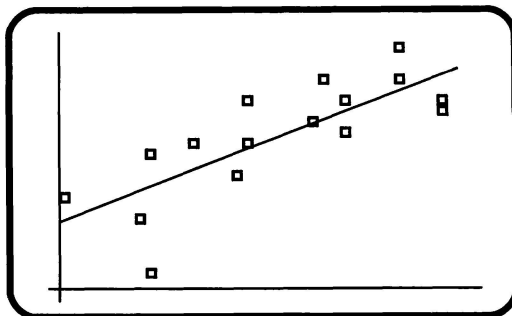
- Use of technology.**

You are encouraged to use

- calculators to solve equations when necessary, to display the terms of a recursively defined sequence when appropriate, and to display graphs,
- spreadsheets on a computer (or calculator),
- the internet for research.

$$\begin{cases} x + y = 1200 \\ 7x + 10y = 9150 \end{cases} \bigg| x, y$$

$$\{x=950, y=250\}$$



|    | A             | B           | C                                                                               | D | E | F | G |
|----|---------------|-------------|---------------------------------------------------------------------------------|---|---|---|---|
| 1  | Physics score | Maths score | <div style="text-align: center;"> <b>Physics and mathematics scores.</b> </div> |   |   |   |   |
| 2  | 6             | 19          |                                                                                 |   |   |   |   |
| 3  | 9             | 12          |                                                                                 |   |   |   |   |
| 4  | 11            | 27          |                                                                                 |   |   |   |   |
| 5  | 12            | 36          |                                                                                 |   |   |   |   |
| 6  | 14            | 42          |                                                                                 |   |   |   |   |
| 7  | 14            | 34          |                                                                                 |   |   |   |   |
| 8  | 16            | 32          |                                                                                 |   |   |   |   |
| 9  | 16            | 27          |                                                                                 |   |   |   |   |
| 10 | 17            | 44          |                                                                                 |   |   |   |   |
| 11 | 19            | 41          |                                                                                 |   |   |   |   |
| 12 |               |             |                                                                                 |   |   |   |   |
| 13 |               |             |                                                                                 |   |   |   |   |
| 14 |               |             |                                                                                 |   |   |   |   |
| 15 |               |             |                                                                                 |   |   |   |   |
| 16 |               |             |                                                                                 |   |   |   |   |
| 17 |               |             |                                                                                 |   |   |   |   |
| 18 |               |             |                                                                                 |   |   |   |   |

Calculators and computers can be particularly useful if we wish to display the terms of sequences. Consider for example the growth in the value of a house that is initially valued at \$500 000 and is subject to an annual increase in value of 6.4%.

$$\text{Initial value} = \$500\,000 \quad \leftarrow T_1$$

$$\text{Value after 1 year} = \$500\,000 \times 1.064 \quad \leftarrow T_2$$

$$\text{Value after 2 years} = \$500\,000 \times 1.064^2 \quad \leftarrow T_3$$

$$\text{Value after 3 years} = \$500\,000 \times 1.064^3 \quad \leftarrow T_4 \quad \text{etc.}$$

These values form a geometric sequence with  $T_{n+1} = T_n \times 1.064$   
and  $T_1 = 500\,000$

We could display the terms of our sequence on a calculator or spreadsheet and view the progressive year by year values, as shown below.

| ◆  | A                    | B | C  | D              |
|----|----------------------|---|----|----------------|
| 1  | Initial value        |   |    | \$500,000.00   |
| 2  | Percentage increase  |   |    | 6.40           |
| 3  | Value at end of year | 1 |    | \$532,000.00   |
| 4  |                      |   | 2  | \$566,048.00   |
| 5  |                      |   | 3  | \$602,275.07   |
| 6  |                      |   | 4  | \$640,820.68   |
| 7  |                      |   | 5  | \$681,833.20   |
| 8  |                      |   | 6  | \$725,470.52   |
| 9  |                      |   | 7  | \$771,900.64   |
| 10 |                      |   | 8  | \$821,302.28   |
| 11 |                      |   | 9  | \$873,865.63   |
| 12 |                      |   | 10 | \$929,793.03   |
| 13 |                      |   | 11 | \$989,299.78   |
| 14 |                      |   | 12 | \$1,052,614.96 |

☒  $a_{n+1} = a_n \cdot 1.064$   
 $a_1 = 500000$   
☐  $b_{n+1} = \square$   
 $b_1 = 0$   
☐  $c_{n+1} = \square$   
 $c_1 = 0$

| n  | a <sub>n</sub> |
|----|----------------|
| 9  | 8.2E+5         |
| 10 | 8.7E+5         |
| 11 | 9.3E+5         |
| 12 | 9.9E+5         |
| 13 | 1.1E+6         |

1052614.96443211

These tables allow us to see, for example, that the value of the house will reach one million dollars shortly after the end of the eleventh year, i.e. early in the 12<sup>th</sup> year.

However note carefully that in this situation, with the recursive definition

$$T_{n+1} = T_n \times 1.064 \quad \text{and} \quad T_1 = 500\,000$$

$T_1$  is the value after **zero** years. Hence we must remember that if we use the ability of a calculator to generate the terms of the sequence, according to the recursive rule given above, then the value after  $n$  years will be given by  $T_{n+1}$ . I.e. in the calculator display above,  $n = 13$  gives the value at the end of 12 years.

One way to avoid this possible source of confusion would be to use the ability of some calculators to accept a sequence defined using  $T_0$  as the 1st term.

I.e. define the sequence as:

$$T_{n+1} = T_n \times 1.064 \quad \text{and} \quad T_0 = 500000,$$

as shown on the right.

Under such a definition  $T_n$  would indeed be the value after  $n$  years.

Alternatively we could use

$$T_{n+1} = T_n \times 1.064 \quad \text{and} \quad T_1 = 500000 \times 1.064$$

and again  $T_n$  would be the value after  $n$  years.

|                                                                                   |        |
|-----------------------------------------------------------------------------------|--------|
| <input checked="" type="checkbox"/> $a_{n+1} = a_n \cdot 1.064$<br>$a_0 = 500000$ |        |
| <input type="checkbox"/> $b_{n+1} = \square$<br>$b_0 = 0$                         |        |
| <input type="checkbox"/> $c_{n+1} = \square$<br>$c_0 = 0$                         |        |
| n                                                                                 | $a_n$  |
| 8                                                                                 | 8.2E+5 |
| 9                                                                                 | 8.7E+5 |
| 10                                                                                | 9.3E+5 |
| 11                                                                                | 9.9E+5 |
| 12                                                                                | 1.1E+6 |
| 1052614.96443211                                                                  |        |

- **The statistical investigation process.**

Given some real world problem, for example estimating the future health needs for a particular region with regards to number of hospital beds required, the statistical investigation process follows the following steps:

- ① Clarify the problem and formulate one or more questions that can be answered with data.
- ② Design and implement a plan to collect and obtain appropriate data.
- ③ Select and apply appropriate graphical or numerical techniques to analyse the data.
- ④ Interpret the results of this analysis and relate the interpretation to the original question; communicate findings in a systematic and concise manner.

(As stated in the Australian Curriculum Glossary for General Mathematics and the West Australian Glossary for Mathematics Applications.)

- **Algorithms.**

Remember, an algorithm is a defined set of steps that that can be applied and systematically followed to solve a particular type of problem. In unit three of *Mathematics Applications* we encountered an algorithm for determining the shortest path between two vertices in a network.